

PII SENTINEL: ENTERPRISE DSPM + DLP

Executive Product Guide, Architecture Deep-Dive & PPT Companion

PRESENTATION BRIEF: This executive briefing guide is structured specifically to prepare you for tomorrow's presentation. It breaks down the product's architecture, security fundamentals, live demo flow, and slide talking points.

1. The 60-Second Executive Pitch

PII Sentinel is an air-gapped, zero-cloud Windows endpoint security platform combining Data Security Posture Management (DSPM) and real-time Data Loss Prevention (DLP). It discovers sensitive unstructured data across enterprise drives and actively blocks accidental leaks in Microsoft Office before data touches the disk.

Core Capabilities:

- **At-Rest Discovery:** Multi-threaded crawl of folders and physical drives with 36 PII and secret recognizers.
- **Real-Time SaveGuard:** In-memory interception in Word and Excel that halts unapproved saves with CISO overrides.
- **Automated Remediation:** Format-aware visual & NTFS watermarking backed by 100% byte-for-byte rollback guarantees.

2. The Problem We Solve

- **Uncontrolled Data Sprawl:** Thousands of unmanaged documents reside on employee laptops containing PAN cards, Aadhaar numbers, customer emails, and server credentials.
- **Severe Regulatory Penalties:** Fines under India's DPDP Act 2023 reach up to ₹250 Crores per violation. GDPR imposes fines up to 4% of global turnover.
- **Cloud DLP Privacy Risk:** Traditional scanners require uploading files to third-party clouds, inadvertently creating new exposure vectors.

3. Technical Architecture & Component Matrix

Component	Technology	Key Enterprise Benefit
Desktop GUI	PySide6 / Qt	High-contrast Dark/Light theme, Oms main thread blocking, responsive telemetry deck.
Extraction	Apache Tika + OCR	Universal text parsing across DOCX, XLSX, PPTX, PDF, and scanned images.
Detection	Microsoft Presidio	Combines spaCy NLP models with Verhoeff-validated Aadhaar, PAN, and cloud secrets.
Office Add-in	C# VSTO / COM	Native pre-save block in Word & Excel; runs without Administrator privileges.
Microservice	FastAPI (127.0.0.1)	Local loopback API secured by bearer token and Windows DPAPI encryption.
Storage	SQLite WAL Mode	High-concurrency database writes eliminating locked-database contention.

4. Slide-by-Slide PPT Blueprint

- Slide 1: Vision:** PII Sentinel — Air-Gapped Enterprise DSPM + DLP Desktop Security
- Slide 2: The Crisis:** The Hidden Danger on Corporate Laptops & Massive Regulatory Fines
- Slide 3: Dual Defense:** Forensic Scanning at Rest + Active In-Memory Pre-Save Protection
- Slide 4: Detection AI:** 36 Global, Indian (Aadhaar/PAN), and Secret Recognizers + Purview Tiers
- Slide 5: Live DLP:** Office Word & Excel Pre-Save Interception with Mandatory Override Rationale
- Slide 6: Remediation:** Native Visual & ADS Watermarks with Automated Byte-for-Byte Rollback
- Slide 7: Trust & Sec:** 100% Offline, Zero Telemetry, Windows DPAPI Encryption, and Token Auth
- Slide 8: Live Demo:** Scanning, Flagging, Watermarking, and Office Pre-Save Interception
- Slide 9: ROI & Impact:** Instant Compliance Audit Readiness with Zero Server Infrastructure
- Slide 10: Conclusion:** Summary & Audience Q&A; Session

5. Live Demo Quick Guide (What to Click)

- Open App:** Highlight Dark/Light theme and the '100% OFFLINE' badge.
- Run Scan:** Select a target folder and start scan. Point out the live telemetry KPI deck.
- Inspect Findings:** Show Purview badges (Restricted, Highly Confidential) and click 'View Redacted Findings'.
- Apply Watermark:** Open Watermark Review Dialog, apply batch watermark, then demonstrate 'Undo Watermark'.
- Live Interception:** Open Word or Excel, paste an Aadhaar/PAN, and hit Save to show the real-time block modal.

6. Tough Executive FAQs

- Q: Does any document leave the machine?**
A: No. Zero external network calls. 100% air-gapped on local CPU.

Q: Does the Office Add-in require Admin rights?

A: No. It installs cleanly into HKCU without administrative elevation.

Q: What if the protection service crashes?

A: Fail-Closed mode safely prevents document saves until the service is restored.

Q: How are quarantine passwords secured?

A: Encrypted via Windows DPAPI, binding decryption to the active Windows user account.